

Regional Prevalence Mapping of Common Mental Disorders Across UK Regions

Internship Task – ANA-3

NHS Digital – APMS 2014 Dataset

Prepared By –

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1.Introduction:

Mental health problems are increasing in different parts of the UK, so analysing regional mental health data is important for understanding how disorder prevalence changes from one region to another. In this task, I analysed regional mental health prevalence data using the APMS 2014 dataset provided by NHS Digital.

The purpose of this analysis was to compare Common Mental Disorder (CMD) prevalence across different regions of England and create visual representations for better understanding. Python libraries such as Pandas, Matplotlib, and GeoPandas were used for data handling and visualization.

Different graphs and regional intensity maps were created to identify regions with higher and lower prevalence rates.

All adults									
Generalised anxiety disorder	6.7	7.2	5.2	5.3	6.2	6.7	5.9	5.4	5.5
Depressive episode	3.2	4.6	2.7	3.3	3.8	2.6	3.3	2.4	4.2
All phobias	2.3	3.1	2.6	1.3	3.1	1.4	2.5	2.0	3.1
Obsessive compulsive disorder	0.9	0.9	0.8	2.7	1.1	1.2	2.0	0.6	1.5
Panic disorder	0.1	1.3	0.5	0.5	0.4	0.2	0.8	0.3	0.9
CMD - NOS ^b	5.9	8.1	8.3	8.4	9.3	5.6	8.7	5.4	9.8
Any CMD	15.7	19.0	16.7	16.7	18.4	14.4	18.0	13.6	20.9

2. Dataset Description:

The dataset used for this analysis was taken from the Adult Psychiatric Morbidity Survey (APMS) 2014 published by NHS Digital. The dataset contains statistical information related to mental health conditions among adults living in England.

For this task, the regional prevalence table from the APMS dataset was selected. The analysis mainly focused on the age-standardised prevalence values of the “Any CMD” category under the “All adults” section. These values represent the percentage prevalence of Common Mental Disorders (CMD) across different regions of England.

The following regions were included in the analysis:

- North East
- North West
- Yorkshire and The Humber
- East Midlands
- West Midlands
- East of England
- London
- South East
- South West

The selected regional values were later converted into a structured dataframe using Python for visualization and regional mapping analysis.

3. Data Preparation:

After selecting the required regional prevalence values from the APMS 2014 dataset, the data was manually extracted and converted into a structured dataframe using Python. The selected data included age-standardised prevalence percentages of the “Any CMD” category for all adults across different regions of England.

The data preparation process involved organizing the region names and their corresponding prevalence values into a clean tabular format. Python Pandas library was used to create and manage the dataframe.

The following preprocessing steps were performed during this stage:

- Extracted regional prevalence values from the dataset
- Assigned region names and CMD prevalence percentages
- Created a structured dataframe using Pandas
- Prepared the dataset for visualization and mapping analysis

After preprocessing, the cleaned dataset was ready for generating graphs and regional intensity maps.

	Region	Any_CMD
0	North East	15.7
1	North West	19.0
2	Yorkshire and The Humber	16.7
3	East Midlands	16.7
4	West Midlands	18.4
5	East of England	14.4
6	London	18.0
7	South East	13.6
8	South West	20.9

4. Data Visualization:

After preprocessing the dataset, different visualization techniques were used to analyse the regional prevalence of Common Mental Disorders across England. Visualization helps in understanding regional differences more clearly and makes the analysis easier to interpret.

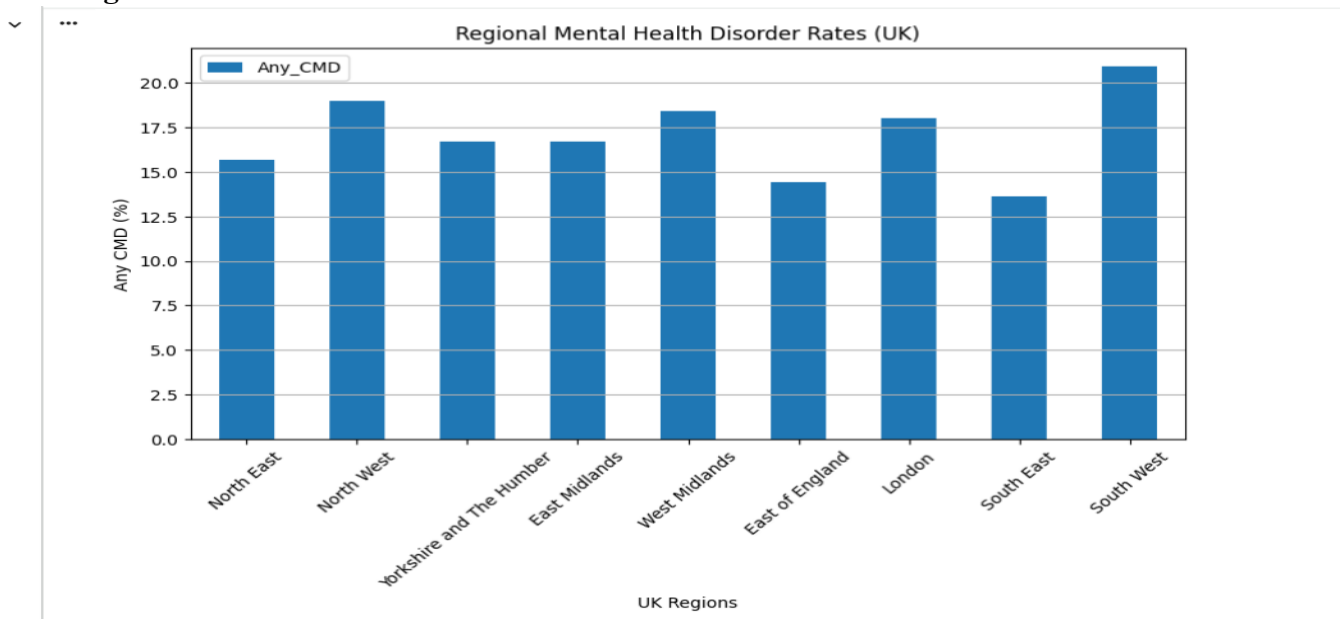
In this task, two major visualizations were created:

- Regional prevalence bar chart
- UK regional intensity map

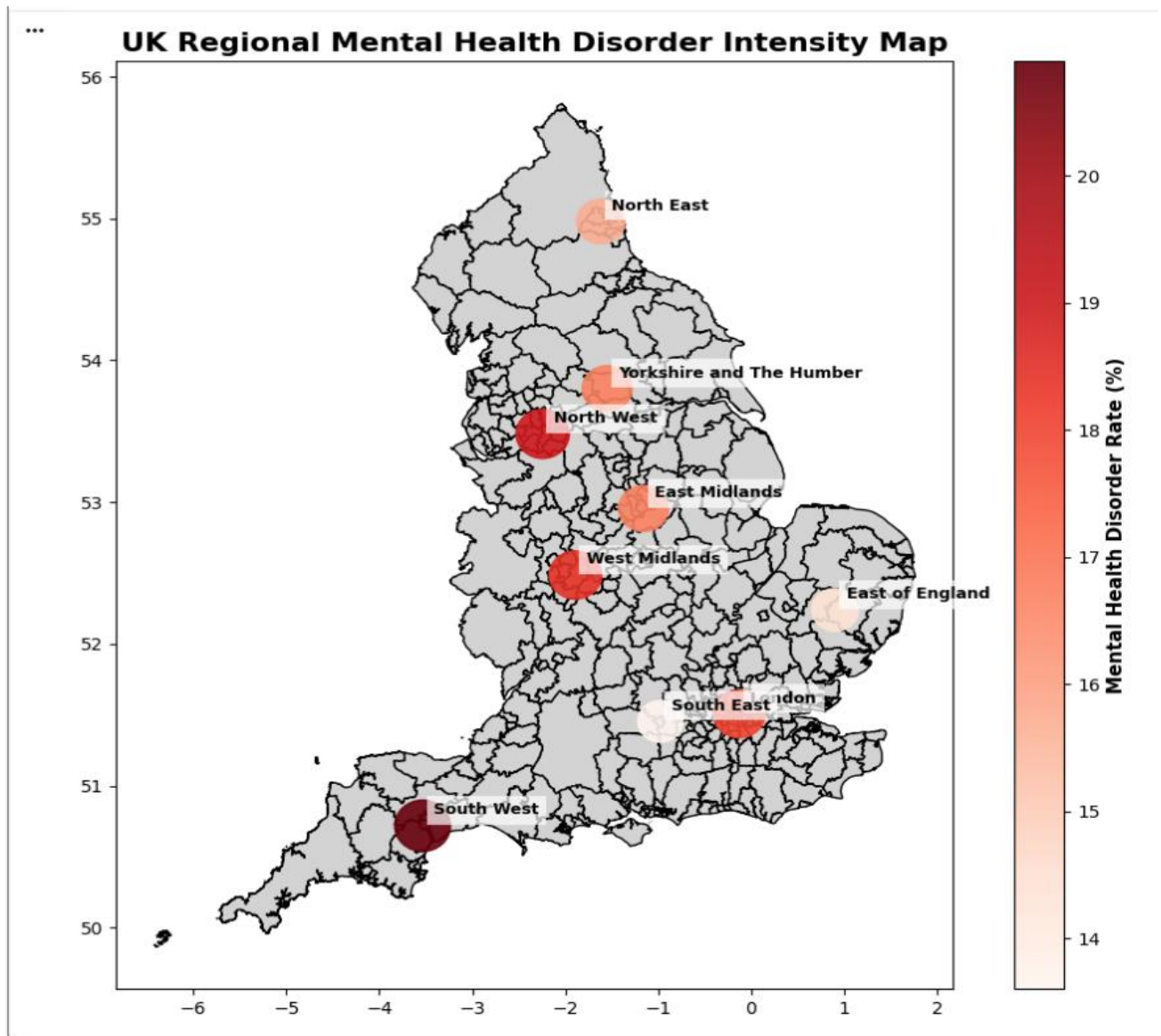
The regional bar chart was used to compare CMD prevalence percentages across different regions, while the regional intensity map was used to visually represent prevalence variation using color intensity and proportional markers.

Python libraries such as Matplotlib and GeoPandas were used to generate these visualizations.

Regional Bar Chart:



UK Regional Intensity Map:



5. Conclusion:

The regional prevalence analysis showed noticeable variation in Common Mental Disorder (CMD) rates across different regions of England. Using the APMS 2014 dataset, regional prevalence values were analysed and visualized using graphs and intensity maps.

The South West region showed the highest CMD prevalence among the analysed regions, while the South East region showed comparatively lower prevalence levels. The visualizations helped in identifying regional mental health patterns more clearly.

Python libraries such as Pandas, Matplotlib, and GeoPandas were successfully used for preprocessing, visualization, and regional mapping analysis. The generated graphs and maps made the regional comparison easier to understand and interpret.

Overall, this analysis provided a better understanding of regional mental health disorder prevalence across England and demonstrated how data visualization techniques can be used for public health analysis.